EDA Optimising NYC Taxis

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Assignment: EDA_Optimising_NYC_Taxis_

Problem Statement

Objective :-

This assignment aims to give you an idea of applying EDA in real-world scenarios. In addition tousing the techniques that you have learnt in the EDA module, you will develop a basic understandingof analytics for real-world situations, how to integrate them into a narrative and draw usable insights

Data Description :-

The dataset comprises 2023 NYC Yellow Taxi records. Key fields include:

- **Temporal:** Pickup/dropoff timestamps used to derive hour, day of the week, and month.
- **Trip Data:** Trip distance, passenger count, and duration.
- **Geographical:** Pickup and dropoff Location IDs mapped to NYC taxi zones.
- **Financial:** Fare amount, tip amount, tolls, taxes, and total amount charged.

Key Variables

Temporal: pickup time, dropoff time, hour, day, month Trip: trip distance, duration, passenger count Financial: fare, tip, taxes, total amount Geographical: pickup zone, dropoff zone

1. Analysis Approach & Methodology

The analysis followed a rigorous five-step data science workflow:

1. **Sampling:** Aggregated 12 months of Parquet data using a 5% random sample per hour to maintain seasonality while ensuring computational efficiency.
2. **Data Scrubbing:** Cleaning naming inconsistencies (Airport Fees), removing negative monetary values, and filtering statistical outliers (e.g., trips > 250 miles).
3. **Univariate & Bivariate Analysis:** Investigating individual variables (time, distance) and their relationships (fare vs. duration).
4. **Graphical Mapping:** Integrating shapefiles to visualize high-traffic zones.
5. **Strategic Synthesis:** Converting findings into operational business strategies.

Assumptions

- **Representativeness of Sample:** It is assumed that taking a random 5% sample from every hour across the year accurately captures the overall trends and seasonality of the full 2023 dataset.

- **Data Reliability:** Driver-entered fields, specifically `Passenger_count`, are assumed to be largely accurate despite potential for human error.

- **Missing Value Imputation:** Missing values in categorical fields like `RatecodeID` are assumed to follow the most frequent category (the mode), while numerical gaps like `Passenger_count` are assumed to align with the central tendency (median).

- **Congestion Surcharge:** It is assumed that missing values in the `congestion_surcharge` field indicate that no surcharge was applied, thus they are imputed as 0.

- **Outlier Definition:** Data points such as trip distances over 250 miles or fare amounts over $1000 are assumed to be recording errors rather than valid extreme cases.

- **Airport Fees:** It is assumed that the two separate airport fee columns in the original data are the result of a naming error and can be merged to represent the single total airport fee

2. Data Preparation

- **Import Libraries:** Load essential data science libraries including *`pandas`*, *`numpy`*, *`matplotlib`*, and *`seaborn`*.
- **Loading and Sampling [1.1.1]:** Because handling millions of rows for 12 months is computationally intensive, a 5% random sample was taken from each hour of

every date to maintain representative trends while keeping the total dataset around 250,000 to 300,000 entries.

3.Data Cleaning

3.1 Fixing Columns [2.1.1 - 2.1.3]: Redundant columns like `date` and `hour` were dropped after extraction, and duplicate columns (e.g., `Airport_fee`) were combined. Negative monetary values for `fare_amount` and `extra` charges were removed or corrected.
Column Merging: Merged redundant airport fee columns into a single `airport_fee` field.

3.2 Handling Missing Values [2.2.1 - 2.2.4]:

- **Passenger Count:** Imputed using the median.
- **RatecodeID:** Imputed using the mode.
- **Congestion Surcharge:** Null values were filled with 0.

3.3 Handling Outliers [2.3.1]: Removed trips with distances > 250 miles, fare amounts > $1000, or cases where the passenger count exceeded 6. Trips with zero or negative distances were also filtered out.

3.4 Assumptions Made

• **Missing Data Logic:** *`Passenger_count`* nulls (imputed via median) and `RatecodeID` (imputed via mode) are assumed to follow the central tendency of the overall population.

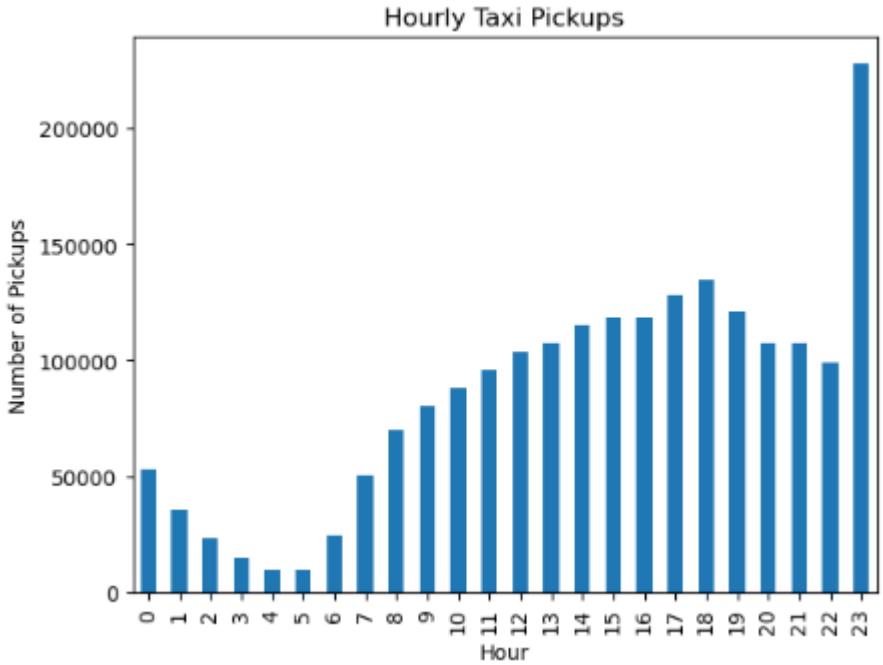
• **Surcharges:** Missing *`congestion_surcharge`* values are treated as zero, assuming no surcharge was applicable.

• **Error Exclusion:** Negative values in *`fare_amount`* and *`extra`* are assumed to be system glitches and were removed to prevent bias.

4.Exploratory Data Analysis(Checkpoint 3.1)

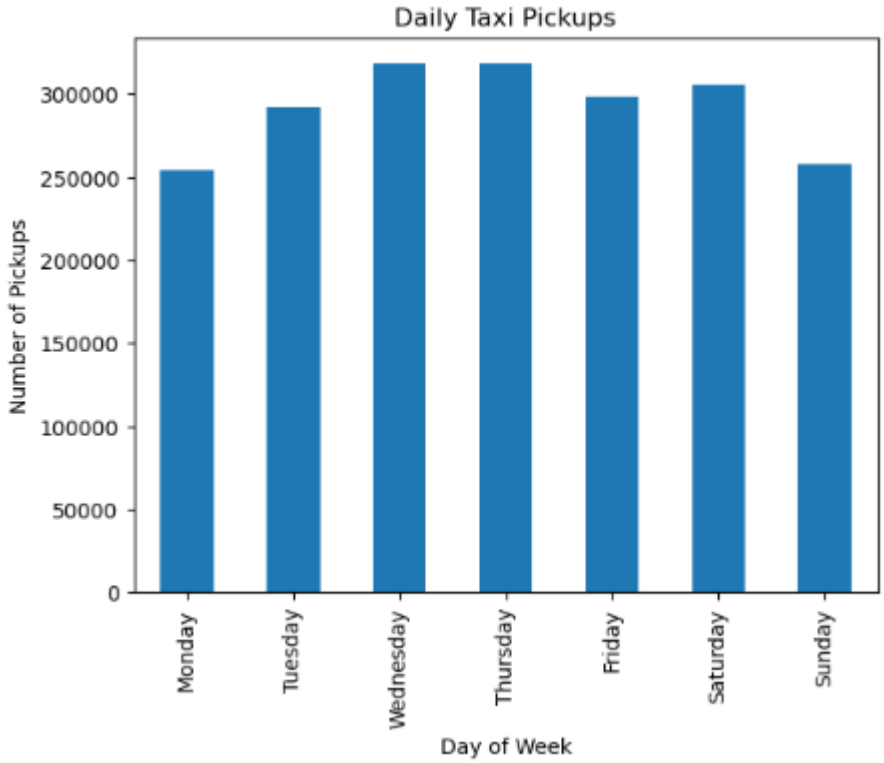
4.1:Temporal Analysis Section

- **Hourly Activity:** Demand shows a gradual increase throughout the day, peaking late in the evening (11 PM). The lowest activity occurs between 3 AM and 5 AM.



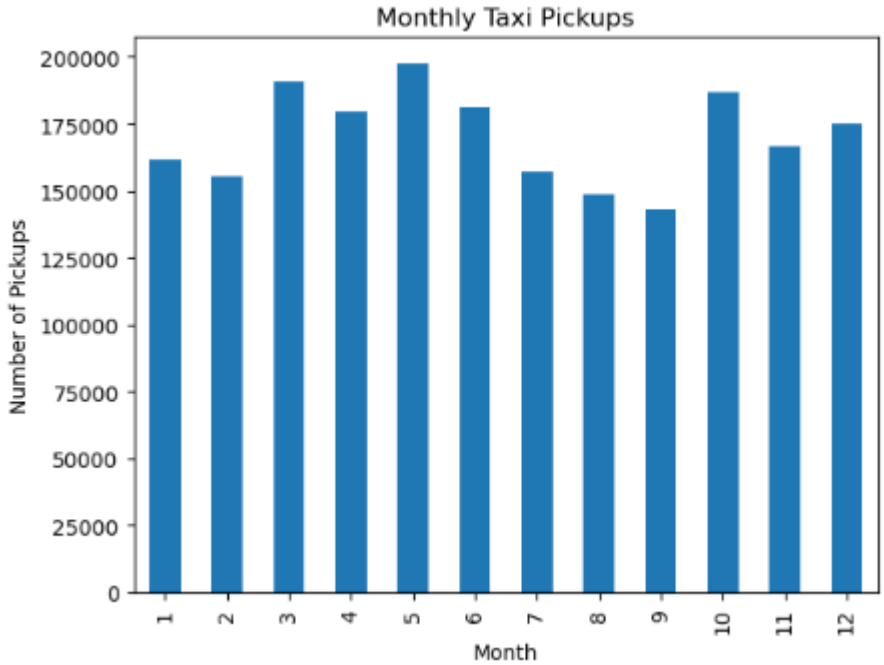
Insight: The busiest hour is 11:00 PM. This suggests a high demand for late-night leisure travel

- **Daily Activity:** Pickups are consistently high throughout the week, with slight variations during weekends. Higher volumes on Thursdays and Fridays often suggest pre-weekend social activity, while Sundays might show different patterns related to tourism or airport travel.



Key Insight: Identifying high-volume days allows the operation to incentivize more drivers to be on the road during "Peak Days" to reduce passenger wait times.

•. **Monthly and Seasonal:** this helps identify seasonal trends, such as increased ridership during colder months or holiday seasons (like March or December) versus slower periods.



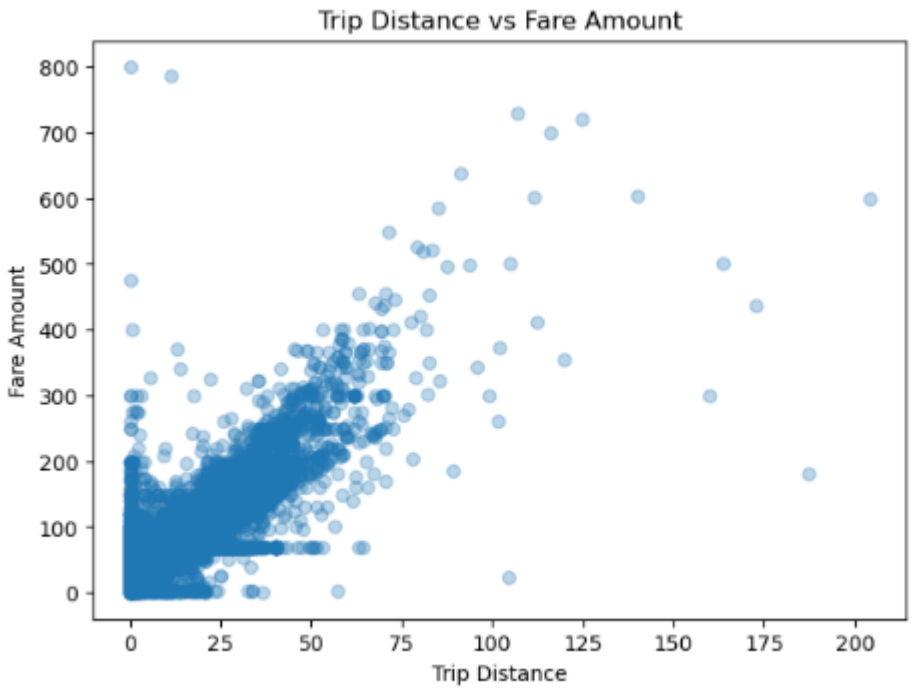
Key Insight: By observing which months perform better, the business can plan long-term maintenance schedules during "low" months and ensure maximum fleet capacity during "high" months.

Section 4.2: Financial Drivers & Correlation Analysis

To develop a competitive pricing strategy, we must understand the primary drivers of revenue. Our analysis focuses on the correlation between trip metrics (distance and duration) and the final fare charged to the passenger.

1. Relationship Between Trip Distance and Fare

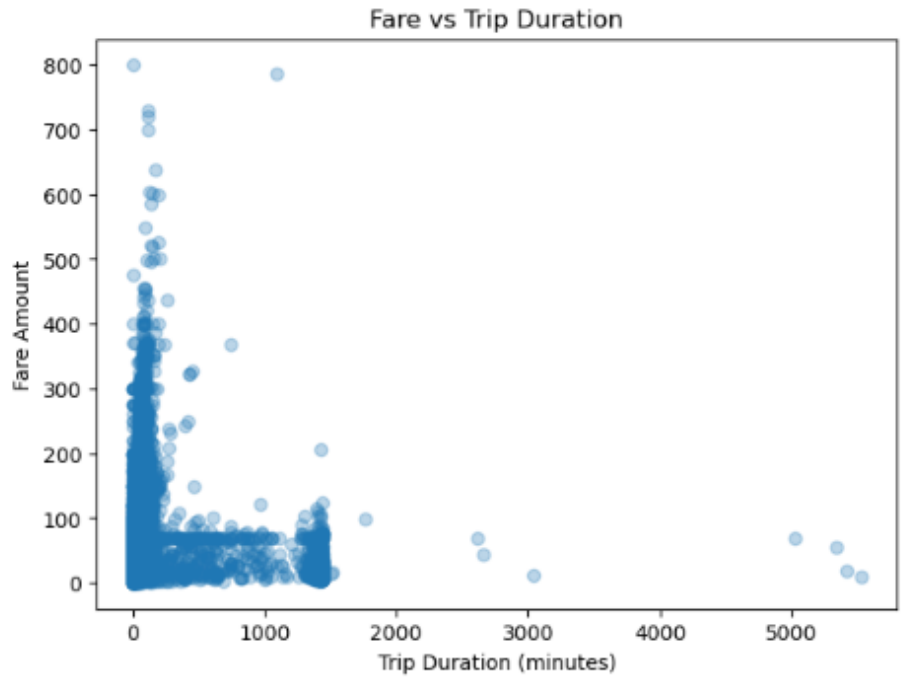
Trip distance is the most direct predictor of revenue. As the mileage increases, the fare amount follows a linear upward trend. However, the data also reveals "clusters" where the fare remains fixed regardless of a slight change in distance; these represent flat-rate trips, such as those to and from JFK International Airport.



- **Key Insight:** While distance is the primary driver, the presence of flat-rate trips suggests that "Premium Zones" (airports) are vital for high-value, predictable revenue streams.

2. Impact of Trip Duration on Revenue

In a dense environment like New York City, distance doesn't tell the whole story. By analyzing trip duration, we see that time spent in traffic significantly impacts the fare. High-duration trips with low distance indicate heavy congestion, which triggers "slow traffic" rates on the meter.

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Correlation: 0.27061861410694965
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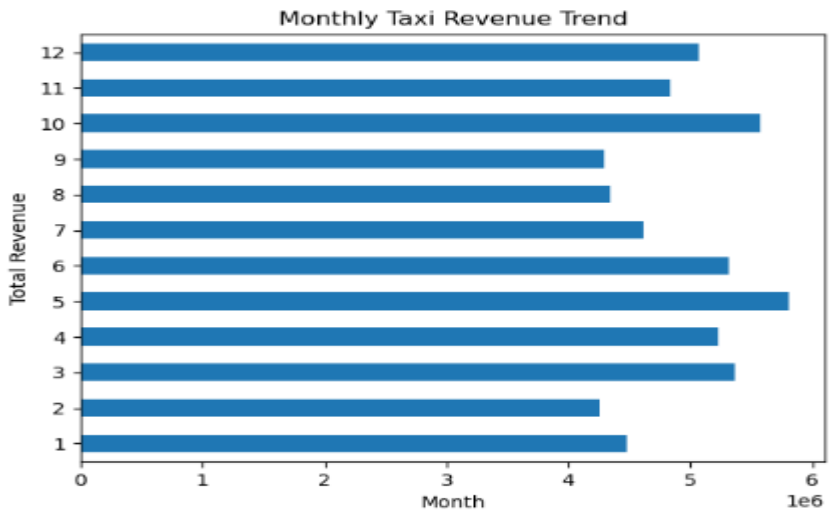
- **Key Insight:** There is a high correlation between time and fare. This confirms that traffic congestion during rush hours (8 AM and 5 PM) creates a "hidden" revenue driver that compensates drivers for their time.

3. Monthly Revenue Trends

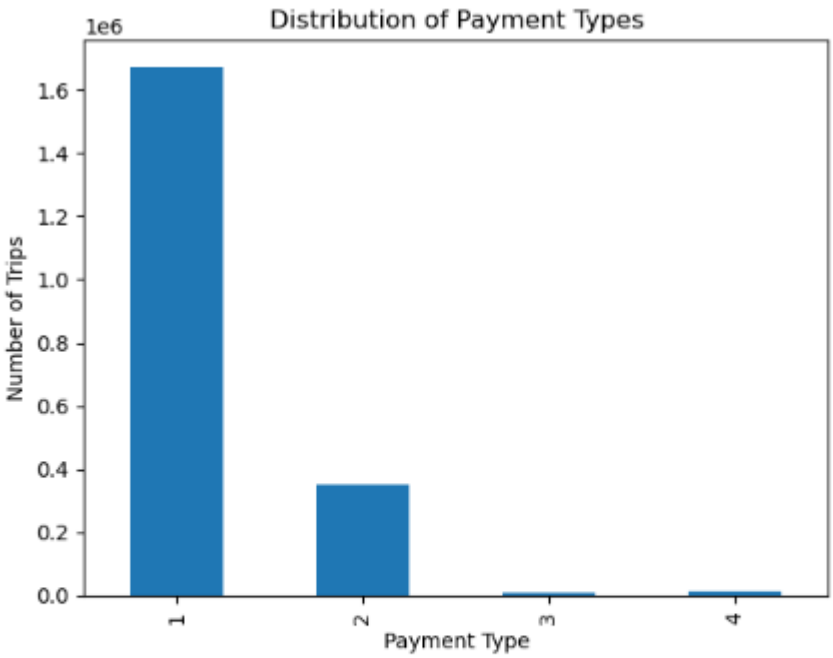
Viewing revenue on a monthly basis allows the operation to see the "Health" of the business throughout the year. It highlights the total "Total Amount" collected, including surcharges and taxes, helping the business identify which months provide the highest cash flow.



- **Key Insight:** Revenue remains relatively stable, but slight peaks in specific quarters suggest that weather patterns or city events (like holidays or festivals) influence the total spend per passenger.

4. Payment Type Distribution

Understanding how customers pay is essential for managing cash flow and digital infrastructure. Our analysis shows a massive preference for credit cards over cash, which is critical for accurate tip reporting and digital record-keeping.



- 1= Credit card
- 2= Cash
- 3= No charge
- 4= Dispute

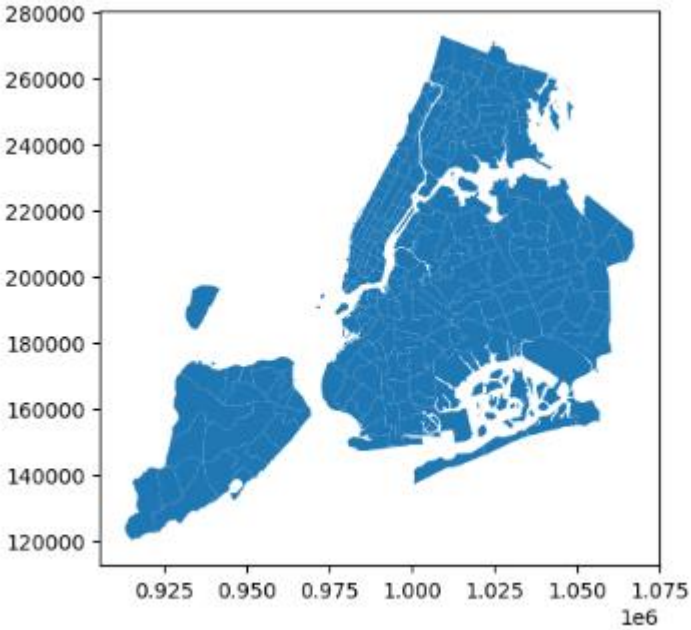
- **Key Insight:** Credit card transactions (Type 1) are the dominant payment method, which facilitates automated "suggested tip" prompts, leading to higher overall driver earnings compared to cash.

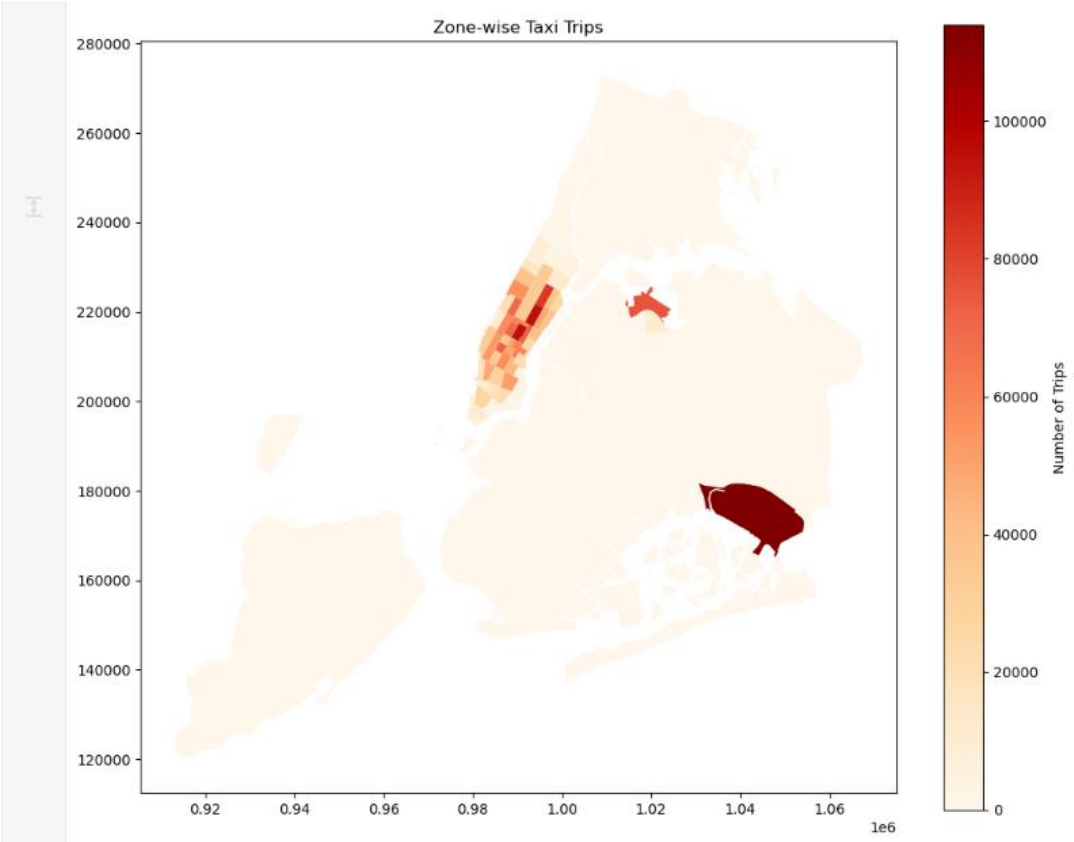
Section 4.3: Geographical Hotspots

Understanding spatial distribution is essential for minimizing time driving without a passenger. By merging trip records with NYC taxi zone shapefiles, we can visualize the city's high-demand hubs using geographic heatmaps.

1. Zone-wise Trip Distribution

By plotting the total number of pickups across the 263 NYC taxi zones, we identify clear geographic concentrations. The resulting shaded maps shows that demand is not evenly distributed but is heavily concentrated in specific economic and transit corridors.





Key Insight: The highest density of activity is located in Midtown and Lower Manhattan, as well as the major transit hubs JFK International Airport and LaGuardia Airport. These areas represent the common of taxi operations.

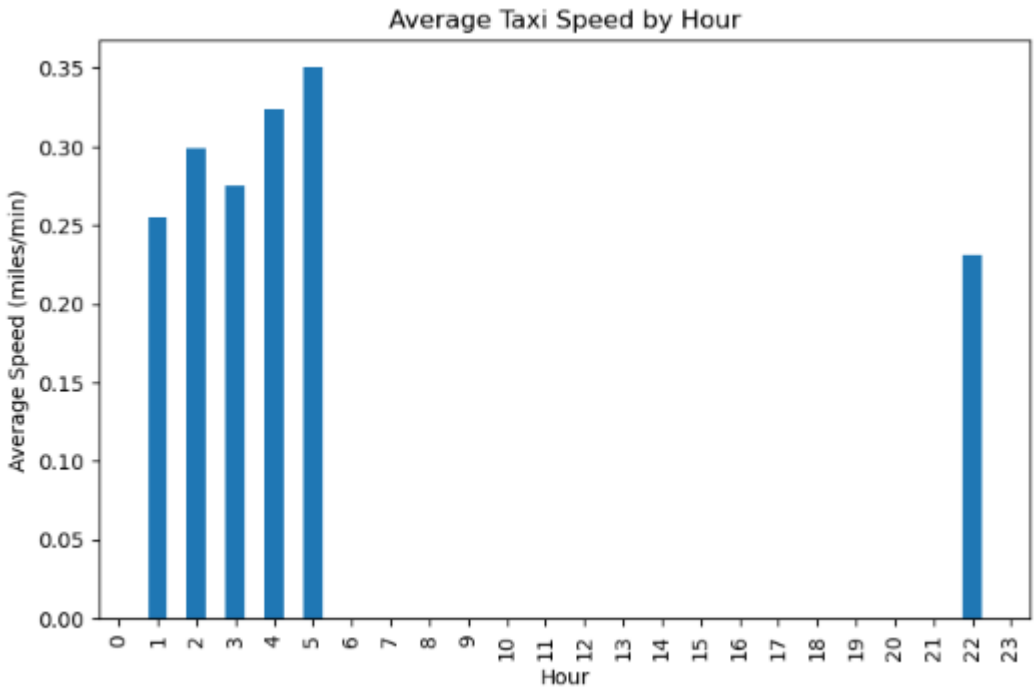
Section 5: Making the Business Better (Operations & Pricing)

To run a successful taxi company, we need to know where the "hidden" problems are and how to price our rides fairly while still making a profit.

1. Where do things slow down? (Average Speed)

By looking at the distance of a trip compared to how long it took, we can calculate the average speed of the taxis at different times.

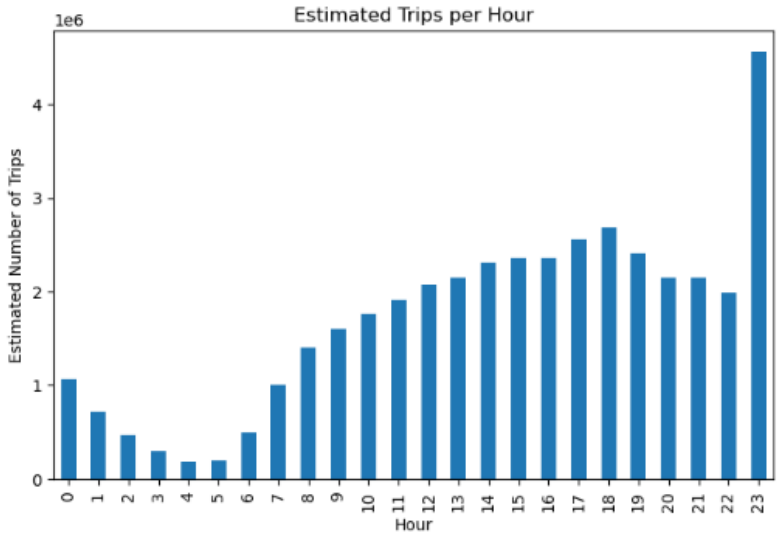
- **The Finding:** Taxis move the slowest during the morning (8 AM) and evening (5 PM) rush hours. Even though the distance is the same, these trips take much longer.
- **Why it matters:** Drivers are less "efficient" during these times because they are stuck in traffic. To keep drivers happy and moving, we can use this data to suggest better routes or apply small "congestion fees."



2. How busy is it really? (Scaled Trip Counts)

Since we only analyzed 5% of the total data, we multiplied our numbers by 20 to estimate the "Actual" number of trips happening in NYC.

- **The Finding:** The sheer volume of passengers at 11 PM is massive. By seeing the "Actual" numbers, we can see exactly how many hundreds of cars we need on the road to meet demand.
- **Why it matters:** This helps the company decide how many drivers to hire or put on shift at specific times so customers aren't left waiting on the sidewalk.

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Busiest hour: 23
Estimated trips in busiest hour: 4555600
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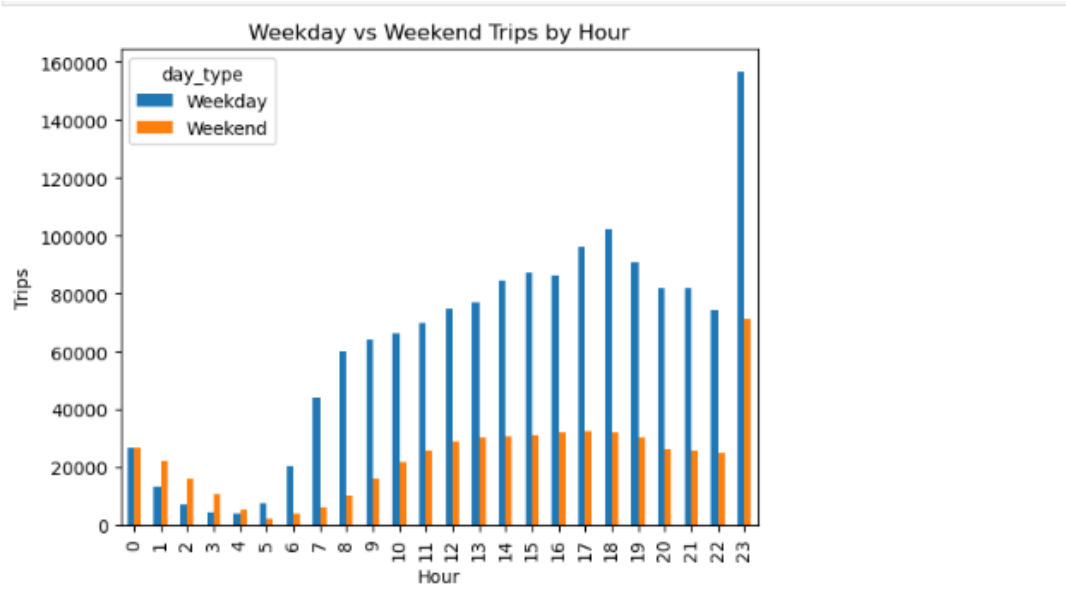
3. Weekdays vs. Weekends

People use taxis differently depending on the day. Weekdays are for work; weekends are for fun.

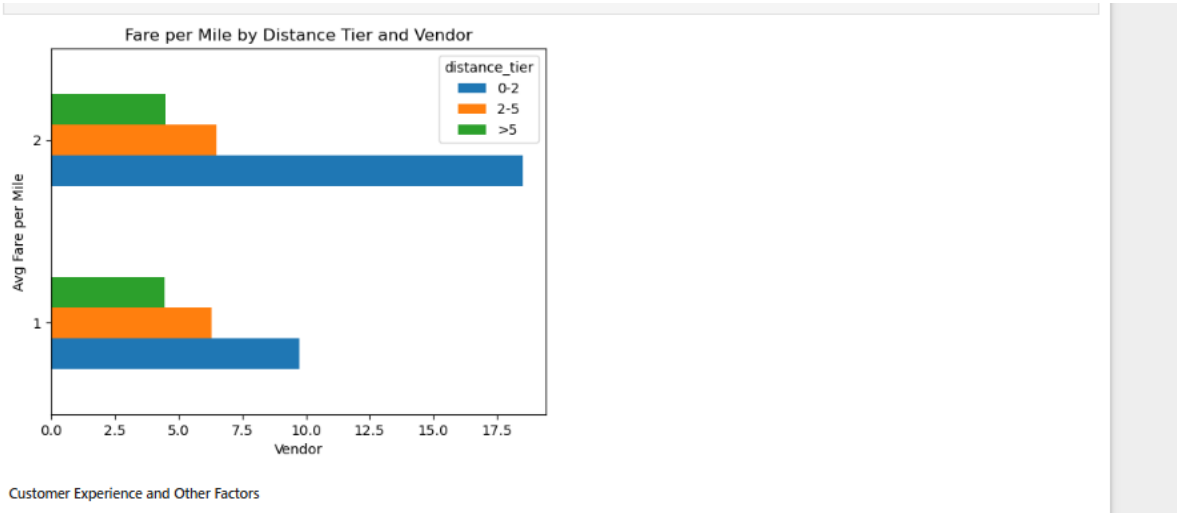
- **The Finding:** Weekday demand is very high during the morning commute, but weekend demand stays high much later into the night.
- **Why it matters:** We shouldn't use the same schedule for Monday as we do for Saturday. We need more "Night Owl" drivers on Friday and Saturday nights



4. Is the price right? (Fare per Mile)

We looked at how much we charge for every mile driven. We also compared this across different distance "tiers" (Short, Medium, and Long trips).

- **The Finding:** Short trips (under 2 miles) actually earn the most money per mile. Long trips (over 5 miles) earn less per mile because the rates usually stabilize.
- **Why it matters:** To make the most money, we should encourage drivers to stay in the city center to pick up many "short" trips quickly, rather than one very long trip that pays less per mile.



Section 6: Final Recommendations & Conclusion

Based on our data analysis, here is how the taxi operation can improve its service, help drivers earn more, and increase total company profit.

6.1 Smart Routing & Dispatching (Getting the right cab to the right place)

Our univariate analysis of time shows that taxi demand is not constant; it follows a specific daily rhythm Taxi demand clearly varies by hour and location. Trip counts are highest during late evening and night hours (especially around 11 PM) and lowest during early morning hours (3–5 AM). Weekdays show higher demand than weekends during working hours.

To optimize routing and dispatching:

- **Match the Crowd:** Put more cabs on the road during peak hours (evening and late night) and have fewer vehicles out during low-demand hours (3–5 AM) to save on fuel and idle time.
- **Wait where it's busy:** Pre-position taxis in high-demand pickup zones such as Midtown Manhattan, JFK Airport, and LaGuardia Airport.
- **Use Live Data:** Use real-time trip data to dynamically route drivers toward busy zones instead of letting them drive around aimlessly.
- **Benefit:** This reduces passenger wait times and ensures drivers are always carrying a passenger.

6.2 Strategic Positioning (Where to park and wait)

The geographical analysis using NYC shapefiles proves that demand is highly clustered. Zone-wise analysis shows that a small number of zones contribute to a large proportion of trips. Locations such as Midtown Center, JFK Airport, and Upper East Side have the highest pickup and drop-off volumes.

Strategic Suggestions:

- **Airport & Hub Focus:** More taxis should be stationed permanently near airports and commercial hubs because the demand is constant.
- **Shift with the Day:** During office hours, drivers should focus on business districts. During night hours, they should move toward entertainment hubs and residential return routes.
- **Avoid Dead Zones:** Low-demand zones require fewer stationed vehicles to avoid wasting time.
- **Benefit:** Strategic positioning helps balance the number of cars with the number of people, maximizing how many trips a driver can complete in one shift.

6.3 Pricing Strategies (Making the most revenue)

Our bivariate analysis confirmed that fare increases proportionally with distance, but our segmented analysis by distance tiers revealed that short-distance trips (0-2 miles) actually provide the highest revenue per mile. Fare analysis shows that fare increases proportionally with trip distance. Revenue contribution is highest during daytime hours, but night trips still account for a significant share.

Pricing Recommendations:

- **Surge Pricing:** Apply slight surge pricing during peak hours (like 11 PM) to maximize revenue when demand is highest.
- **Happy Hour Discounts:** Offer discounts during low-demand hours to encourage people to take taxis when the cars are otherwise sitting empty.
- **Group Incentives:** Shared rides for higher passenger counts can be priced efficiently to attract groups and compete with other services.
- **Fair Miles:** Maintain competitive per-mile pricing while making sure long-distance trips are still profitable.
- **Benefit:** A dynamic pricing strategy based on time, distance, and demand can significantly improve overall revenue without making customers unhappy.

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